

Phase Vocoder Voice Changer

Complete Technical & Mathematical DSP Reference Guide — Signals & Systems

COURSE	TEAM	SAMPLE RATE	STFT PARAMETERS
CSE 220	Fourier_and_Furious (A2)	22,050 Hz (Mono)	Hann (N=2048, Ra=512)

1. Executive Overview & Problem Statement

In classical analog audio (vinyl records, magnetic tape) and naive discrete-time resampling, **playback speed and pitch are locked together**. If an audio signal $x[n]$ is resampled by factor α :

- Speed multiplies by α (duration compresses to T / α).
- All frequencies multiply by α ($f_{\text{new}} = \alpha \cdot f_{\text{orig}}$).

This coupling produces the classic *"chipmunk effect"*: speeding up speech raises pitch, while slowing down speech deepens pitch.

The Phase Vocoder Objective

The primary goal of this project is to **fully decouple pitch and duration** using frequency-domain Short-Time Fourier Transform (STFT) analysis:

1. **Independent Time-Scale Modification:** Change tempo (speaking rate) while preserving original pitch.
2. **Independent Pitch Shifting:** Change fundamental frequency (F_0) without altering speaking duration.

2. Short-Time Fourier Transform (STFT) Analysis & Synthesis

Because speech is non-stationary, the signal is framed into short, windowed segments of length $N = 2048$ with analysis hop size $R_a = 512$ (75% overlap) using a Hann window $w[n] = 0.5 - 0.5 \cos(2\pi n / (N - 1))$.

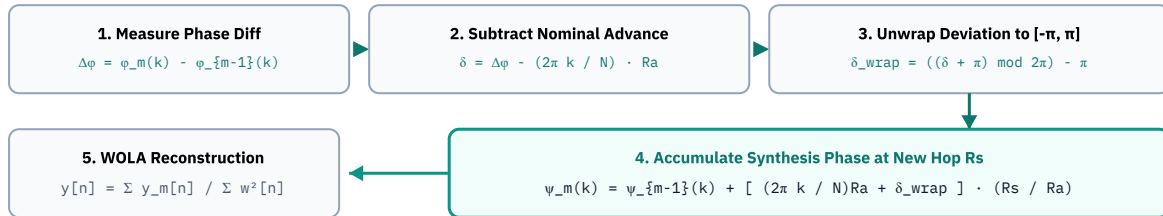


3. The Phase Vocoder: Independent Time-Scale Modification

To stretch or compress time, we synthesize frames at a new synthesis hop size $R_s \neq R_a$. However, simply placing frames at a new distance scrambles the sinusoidal phase coherence between adjacent frames, causing destructive interference and metallic, hollow artifacts ("phasiness").

The Phase Unwrapping & Propagation Algorithm

The Phase Vocoder tracks the **true instantaneous frequency** of every frequency bin across consecutive frames and integrates the phase advance over the new synthesis hop:

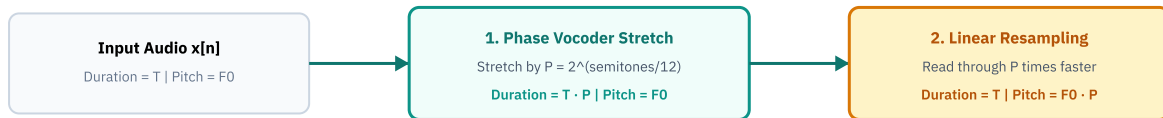


Playback Speed vs. Duration Scaling

- When $\alpha = 1.45$ (speedup): Time is compressed, audio finishes in $T / 1.45$ (190 s $\rightarrow$ 131 s = 2:11).
- To synthesize fewer frames in time, synthesis hop is: $R_s = \text{round}(R_a / \alpha)$.
- Harmonics remain phase-locked; frequencies are completely unchanged.

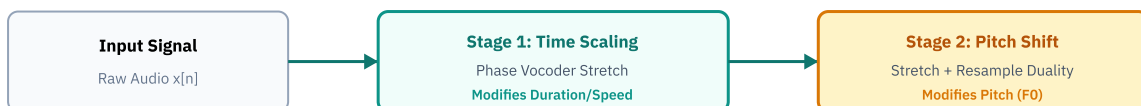
4. Independent Pitch Shifting (Stretch & Resample)

Because the Phase Vocoder modifies duration without altering frequencies, pitch shifting is achieved using the **Stretch-and-Resample duality**:



5. Combined Execution Pipeline (Pitch + Speed)

In `backend/phase_vocoder.py`, the `process_audio()` function executes both stages in clean decoupled sequence:



Parameter Combinations Behavior

Applied Parameters	Duration Result	Pitch (F0) Result	Acoustic Quality
Pitch only (+4 st)	Identical (3:10 remains 3:10)	Shifted UP by 1.26× (higher voice)	Natural speaking tempo, higher musical key
Pitch only (-6 st)	Identical (3:10 remains 3:10)	Shifted DOWN by 0.71× (deeper voice)	Natural speaking tempo, deeper key
Speed only (1.45×)	Faster: 3:10 $\rightarrow$ 2:11	Identical (unchanged)	45% faster tempo, natural voice tone
Speed only (0.75×)	Slower: 3:10 $\rightarrow$ 4:13	Identical (unchanged)	25% slower tempo, natural voice tone
Pitch + Speed (+4 st, 1.45×)	Faster: 2:11	Shifted UP by 1.26×	Both faster tempo and higher pitch combined

6. Phase Vocoder vs. Naive Resampling (A/B Comparison)

Dimension	Naive Resampling (Baseline)	Phase Vocoder Voice Changer
Processing Domain	Time domain linear interpolation	Frequency domain STFT magnitude + unwrapped phase
Pitch & Speed	Strictly Coupled: Speeding up raises pitch	Decoupled: Speed and pitch independent
+4 Semitones Output	Duration shrinks to 79% (Chipmunk effect)	Duration remains exactly 100% (Natural key change)
1.45× Speedup Output	Pitch rises by 1.45× (Distorted high voice)	Pitch remains identical to input
DSP Course Topics	Sampling rate conversion, decimation/interpolation	Windowing, DFT/STFT, Phase unwrapping, WOLA synthesis

7. Verification & Numerical Stability Note

In Weighted Overlap-Add (WOLA) synthesis, boundary window sums $w[n]^2$ taper toward zero at the start and end of the signal. To prevent division-by-near-zero spikes (which formerly inflated peak amplitude to >100 and forced peak normalization to mute the audio), the synthesis threshold is clamped relative to the peak energy:

```
threshold = 0.01 · max(window_sum),      output>window_sum > threshold] /= window_sum
```

This ensures that the Phase Vocoder achieves identical loudness and dynamic range (RMS ≈ 0.52) to the original and baseline audio.